Errors-in-Variables Models

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Variables Measured Without Error

Suppose we have a simple linear regression with response y and predictor x:

$$\eta = b_0 + b_1 x$$

$$y \sim N(\eta, \sigma)$$

This model implicitely assumes that x and y are measured without error

This assumption is so common that most people are not even aware of it

Predictor Variables Measured With Error

Suppose we have a simple linear regression with response y and *observed* predictor x with known measurement error s_x.

1. Model x as measuring a true *latent* variable x_L:

$$x \sim N(x_L, s_x)$$

2. Define a hierachical prior on x_L:

$$x_L \sim N(\mu_L, \sigma_L)$$

3. Use x_L as a predictor of y:

$$\eta = b_0 + b_1 x_L$$

$$y \sim N(\eta, \sigma)$$

This is called an errors-in-variables (EiV) model

Response Variables Measured With Error

Suppose that y was measured with error with known measurement error s_y:

1. Model y as measuring a true *latent* variable y_L:

$$y \sim N(y_L, s_y)$$

2. Predict y_L using the predictor term η (whatever it contains):

$$y_L \sim N(\eta, \sigma)$$

brms Syntax for EiV Models

Noise in predictor x with measurement error s_x:

```
y ~ me(x, sx)
```

Noise in response y with measurement error s_y:

```
y | mi(sy) ~ x
```

Noise in both variables:

```
y | mi(sy) ~ me(x, sx)
```

Assumptions of EiV models

Error made by measurement is Gaussian:

$$x \sim N(x_L, s_x)$$

Measurement error s_x is known (without error)

Latent variables are uncorrelated by default:

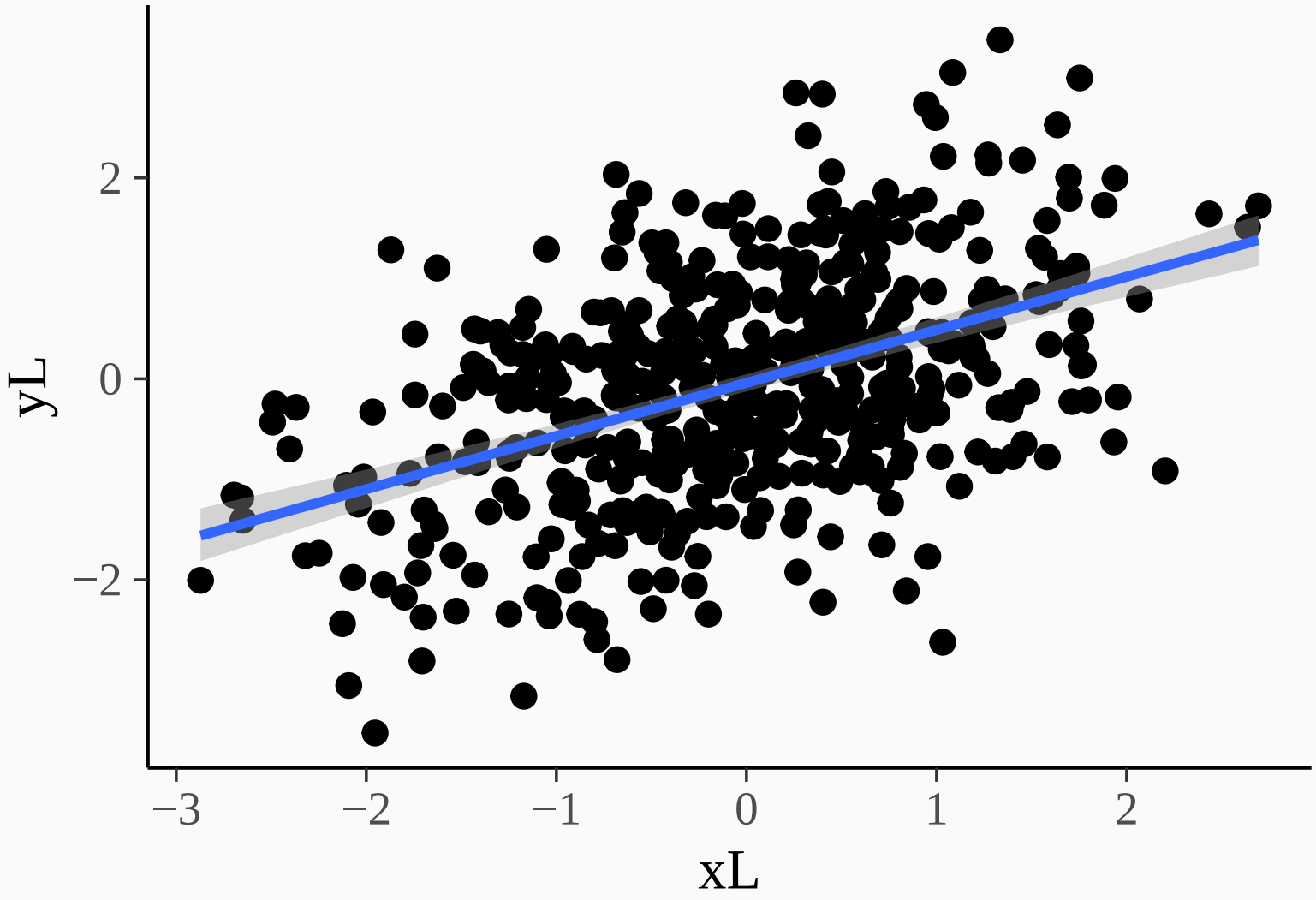
$$\text{cor}(x_{L1}, x_{L2}) = 0$$

Model correlations between latent variables in brms via:

```
bf(y ~ me(x1, sx1) + me(x2, sx2)) + set_mecor(TRUE)
```

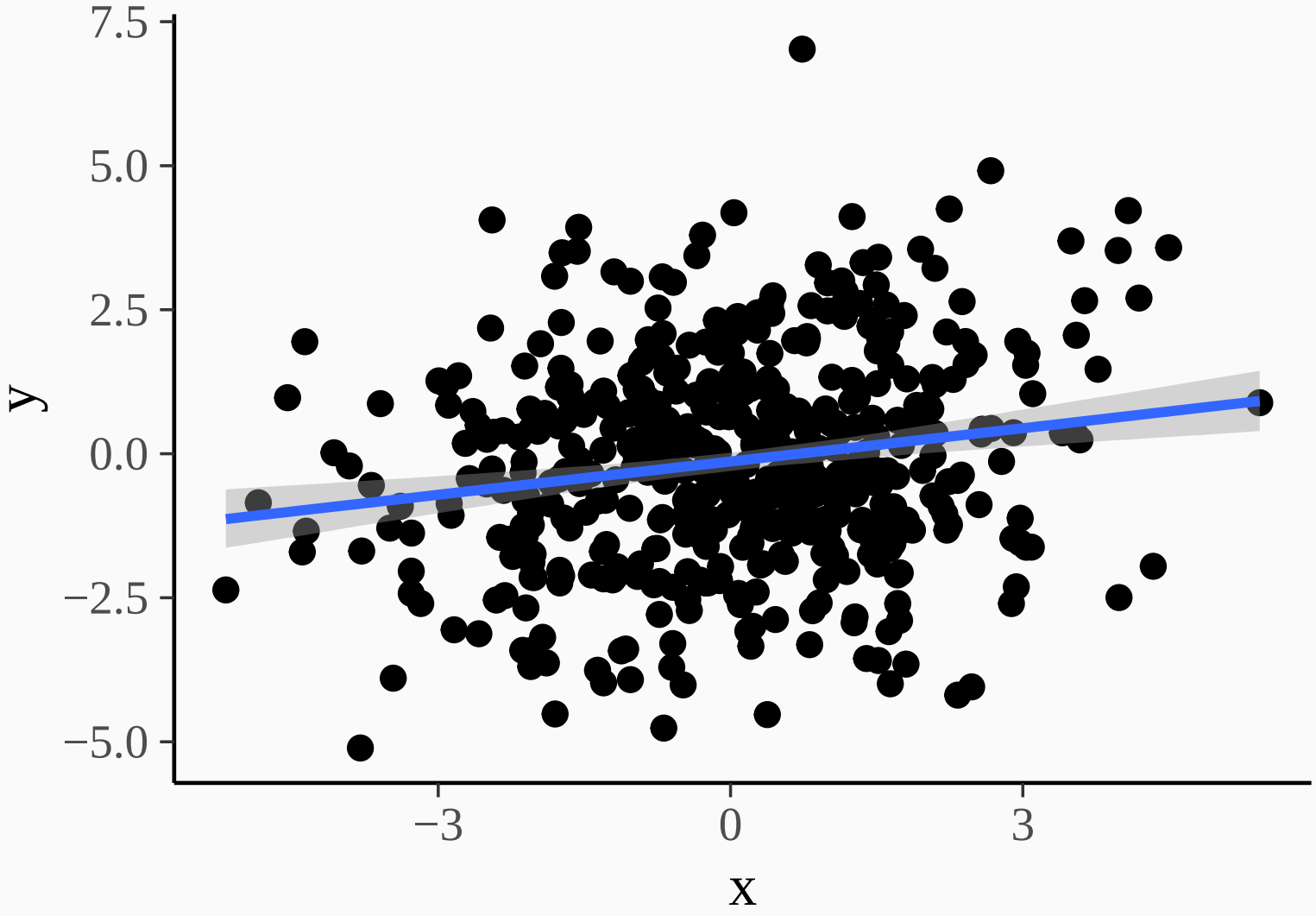
Example: True Relationship

```
yL ~ xL
```



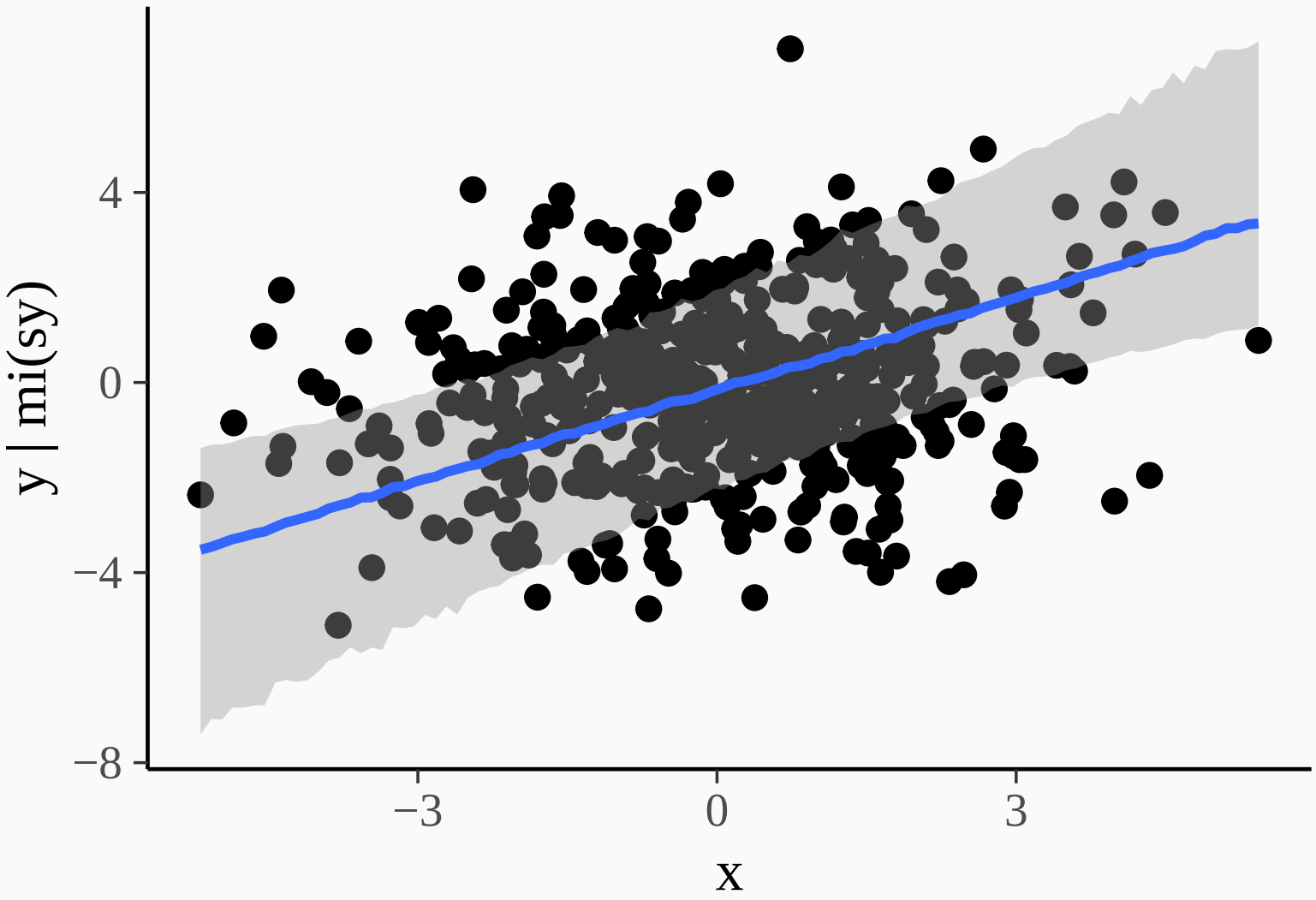
Example: Observed Relationship

```
y ~ x
```



Example: Recovered Relationship

```
y | mi(sy) ~ me(x, sx)
```



Difference to Meta-Analytic Models

In meta-analysis we have obsererved outcomes y with known sampling error s_y.

A random effects meta-analysis assumes heterogeneity of true outcomes y_L.

This implies the following marginal model for y:

$$y \sim N(\eta, \sqrt{s_y^2 + \sigma^2})$$

where σ is the standard deviation of the latent variable y_L.

In meta-analysis, η still predicts y and not y_L as in EiV models

Don't use `y | se(sy) ~ ...` for EiV models!

Time for exercise

'errors_in_variables_models.R'